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RESEARCH

Oral microbiota may predict the presence of esophageal squamous cell carcinoma

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Abstract

Purpose Microbial imbalances have been well elucidated in esophageal adenocarcinoma (EAC), but few studies address the oral microbiota in esophageal squamous cell carcinoma (ESCC). In view of the fact, we aimed to explore the associations of oral microbiota with these patients suffering from ESCC.

Methods In our study, a total of 109 individuals were enrolled (control = 53, ESCC = 56). We profiled the microbiota in oral swabs from individuals with control (ConT) and ESCC (ESCCT). 16S rRNA gene sequencing was applied to analyze the microbiome. The α and β diversity differences were tested by Tukey Test and Partial Least Squares Discriminant Analysis (PLS-DA) respectively. Linear discriminant analysis effect size (LEfSe) analysis was performed to assess taxonomic differences between the two groups.

Results Our results showed that the microbial richness and diversity was a slightly higher in ESCCT groups than that in ConT groups. *Bacteroidota*, *Firmicutes*, *Proteobacteria*, *Fusobacteria*, *Actinobacteria* and *Patescibacteria* were the six dominant bacteria of oral flora in the two groups. When compared with control group, increased *Fusobacterioa* at phylum level, *Neisseriaceae* at family level and *Leptotrichia* at genus level were detected. LEfSe analysis indicated a greater abundance of *Leptotrichiaceae*, *Leptotrichia*, *Fusobacteriales*, *Fusobacteria* and *Fusobacteriota* in ESCC groups.

Conclusion Our study suggests a potential association between oral microbiome dysbiosis and ESCC and provides insights on a potential screening marker for esophageal cancer.

Keywords Esophageal squamous cell carcinoma · Oral microbiota · Tongue coating · Characterization · *Leptotrichia*

Introduction

Esophageal cancer is one of the most common gastrointestinal cancers. The main histological types of esophageal carcinoma are esophageal adenocarcinoma (EAC) and esophageal squamous cell carcinoma (ESCC), and particularly with the latter predominating in the Chinese population (Kelly 2019). The incidence rate of esophageal cancer has been increasing in recent years, but patients with early esophageal cancer may not have obvious symptoms, which make the diagnosis difficult. Although great research progress has been made in the occurrence and treatment of esophageal cancer, there is still no effective treatment, and the survival rate is still very low (Baba et al. 2014; Torre et al. 2015). Therefore, in such a case, elucidation of the pathogenesis underlying ESCC and characterization of novel biomarkers are important and finally new therapeutic alternatives will be offered.

Recently, microorganisms have been linked to a significant number of gastrointestinal disease and several cancers.

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Progress in metagenome-wide association studies of fecal samples has characterized some important microbial markers of CRC (Schwabe and Jobin 2013; Zou et al. 2018), and the causal effect of bacteria on cancer has been recognized (Yu et al. 2017). However, there are few studies on upper gastrointestinal microorganisms, especially esophageal microorganisms and their relationship with related diseases. Previous study (Gao et al. 2016) proved that *Porphyromonas gingivalis* infection was found in cancer tissues and adjacent normal tissues of patients with ESCC, while in contrast to ESCC group, there was no infection in esophageal mucosa of healthy control group, which suggests that the bacteria may play a pathogenic role in ESCC and that the presence of the bacteria is positively correlated with the severity of ESCC, which makes clinical prognosis poor.

Oral cavity is one of the most complex and numerous parts of human microorganisms. When the microbial community and its host are in an equilibrium and steady state, they play physiological barrier roles in the invasion of exogenous pathogens. Contrarily, imbalance can lead to multiple infectious diseases. More importantly, oral microorganisms may play an important role in cancer (Nair et al. 1996) and other chronic diseases (Salaspuro 2003; Karin et al. 2006) through direct metabolism of chemical carcinogens and systemic inflammation. It has also been reported that tooth loss and poor oral hygiene are closely related to upper gastrointestinal cancer, indicating the possible role of oral microorganisms in the occurrence and development of upper gastrointestinal cancer (Abnet et al. 2005, 2008). A study quantified the daily, weekly, and monthly variability of the oral microbiome in healthy individuals only to find that teeth and supra-gingival plaque were more susceptible to oral hygiene habits. On the contrary, within-sample diversity levels were consistent in the tongue and the communities in the salivary and tongue plaque have a high similarity (Mager et al. 2003). In addition, most of the high-abundance operational taxonomic units (OTUs) found in saliva were derived from the tongue.

In the present study, we aimed to assess and compare the diversity and composition of the oral microbiota in human tongue coating between ESCC group and ConT group. Thus, it might further illustrate the role of microbiota in the pathogenesis of ESCC.

Materials and methods

Study participants

We recruited ESCC patients (ESCCT group) and healthy control volunteers (ConT group) in Jinhu County People's Hospital from January to September 2021. All the patients with esophageal cancer were diagnosed as esophageal squamous cell carcinoma by biopsy, while the healthy controls were considered as healthy people without organic diseases, systemic diseases, oral diseases and family history of tumors.

To avoid the impact of food residues on tongue coating collection, we controlled fasting, smoking and alcohol before sampling, meanwhile, we could use a sterile throat swab (Shanghai Xinchao Biotechnology Co., Ltd.) to roll forward along the root of the tongue, brush the tongue coating for 30 times, immerse it in a centrifugal tube (Suzhou Hailu Biotechnology Co., Ltd.) containing phosphate buffer (PBS) (Suzhou Hailu Biotechnology Co., Ltd.) and continuously stir for many times to ensure that microorganisms on the swab are transferred to the buffer as many as possible. In the end, repeat the above steps twice and combine the samples collected for many times, which makes it sure that the collected tongue coating samples are sufficient. When the collected tongue coating samples were centrifuged at 4 ℃ at 5000 g for 5 min, the supernatant was discarded and stored in a refrigerator at −80 ℃. To minimize the potential influence on the microbiota, all the participants enrolled should not receive antibiotics, H_2 receptor antagonists, proton pump inhibitors and probiotics one month before the sample collection. Subjects who had received radiotherapy, chemotherapy, or/and prior surgery were excluded. None of the individuals had active periodontal disease, nasopharyngeal or oropharyngeal cavity illnesses, or dental caries within the two weeks prior to sample collection. Subjects who received dental implants or orthodontic appliances within a year were also excluded. The study protocol was approved by the institutional review board of Nanjing Medical University, and all the experiments were performed in accordance with approved guidelines and regulations.

DNA extraction

Total genome DNA from samples was extracted using the QIAamp DNA Mini Kit (QIAGEN, Valencia, CA, USA) combined with the bead-beating method. The DNA concentrations of each sample were adjusted to 50 ng/μl for subsequent 16S rDNA genes analysis. The bacterial DNA samples were stored at –80 ℃ for sequencing of the follow-up studies.

PCR amplification

16S rDNA genes of V3–V4 region were amplified using universal primers, namely 338F (5′-ACTCCTACGGGA GGCAGCAG-3′) and 806R (5′ -GGACTACHVGGGTWT CTAAT-3′). All PCR reactions (including denaturation, annealing, and elongation) were carried out with Phusion® High-Fidelity PCR Master Mix (New England Biolabs). After electrophoresis of PCR products, samples with bright

main strip between 400 and 450 bp were chosen for the next mixing and purification with Qiagen Gel Extraction Kit (Qiagen, Germany).

Sequencing processing and analysis

Purified amplicons were pooled in equimolar and paired-end sequenced on an Illumina Novaseq6000 PE250 platform (Illumina, San Diego, USA). The raw data were filtered with QIIME (v 1.8.0), discarding the reads which were de-replicated or shorter than 150 bp. Filtered reads were clustered into OTUs with the assumed 97% similarity. Compared with the SILVA database (version 128), the species classification information of each OTU was obtained. All raw reads were stored in NCBI Sequence Read Archive (SRA) database, and the accession number is PRJNA853196.

Statistical analysis

For continuous variables, independent *t* test, White's nonparametric *t* test, and Mann–Whitney *U* test were applied. For categorical variables between groups, Pearson's chi-square or Fisher's exact test was used, depending on assumption validity. QIIME software was used to evaluate the α diversity by calculating the Shannon index and Simpson index. To compare the differences of diversity among groups, β diversity was tested by Partial Least Squares Discriminant Analysis (PLS-DA). Linear discriminant analysis (LDA) effect size (LEfSe) was performed to find key microbes associated with different groups with the LDA threshold of 3. We used Phylogenetic investigation of communities by reconstruction of unobserved states (PICRUSt) analysis to predict Kyoto Encyclopedia of Genes and Genomes (KEGG) biochemical pathways. Statistical analysis was performed using the SPSS V19.0 (SPSS Inc., Chicago, IL) and STAMP V2.1.3. GraphPad Prism version 6.0 (San Diego, CA) was used for preparation of graphs. A p value < 0.05 was considered statistically significant.

Results

Demographic characteristics of all individuals

A total of 53 healthy control volunteers and 56 ESCC patients were enrolled in this study. Demographic characteristics of all included individuals were shown in Table 1. High age was observed in ESCC groups when compared with healthy control volunteers. No significant differences in sex, alcohol intake, smoking, diabetic background and family history of cancer were observed between the two groups.

Table 1 Clinical characteristics of enrolled patients and healthy controls

Characteristics	Control ($n=53$)	ESCC ($n=56$)	p value
Age	49.32 ± 10.43	54.56 ± 10.38	0.032*
BMI (kg/m^2)	24.35 ± 3.56	25.42 ± 4.23	0.189
Sex (male)	26	30	0.773
Smoker (yes)	16	22	0.542
Alcohol consumption (yes)	10	15	0.685
Diabetes (yes)	8	11	0.168
Family history of cancer (yes)	4	7	0.455

*P value is less than 0.05, the difference is statistically significant

Quality control and basic analysis

After sequencing and quality filtering, more than 6.7 million tags and a total of 2629 OTUs were obtained with the dominant length of tags locating among 400–440 bp (Fig. 1A). To test the sequencing depth, we created the rarefaction curves and showed a reasonable amount of sampling (Fig. 1B).

Microbial diversity and richness between the two groups

The microbial α diversity and β diversity were applied to analyze the microbiota biodiversity and composition between the two groups. We used Chao1 index and Shannon index to describe the community richness and diversity. Compared with ConT group, the Shannon index of the ESCCT group increased, indicating a higher microbial diversity ($p=0.33$, Fig. 2A). A slight higher richness of oral microbiota was observed in ESCCT group than in ConT group according to the Chao1 index ($p=0.94$, Fig. 2B). Moreover, the Venn diagram indicated that 1812 of the total 2629 OTUs were shared between the two groups, and 460 and 357 OTUs were unique for ConT and ESCCT group respectively (Fig. 2C). About β diversity, Partial Least Squares Discriminant Analysis (PLS-DA) at the OTU level revealed a statistically significant clustering (Fig. 2D), accounting for different microbial community structures.

The changes of oral microbiota composition between the two groups

As shown in Fig. 3A–D, each group showed a different bacterial composition at the phylum, family, class and genus levels. We explored taxa distribution at the phylum, family, class and genus level to reveal distinctive characteristics of

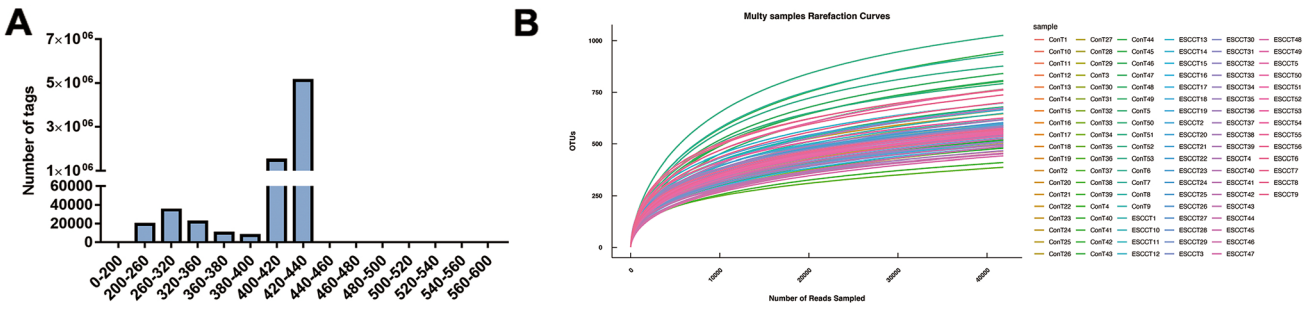


Fig. 1 Quality Control and Basic Analysis. **A** The abscissa is the sequence length of tags, and the ordinate is the number of tags. **B** Rarefaction curves for OTUs

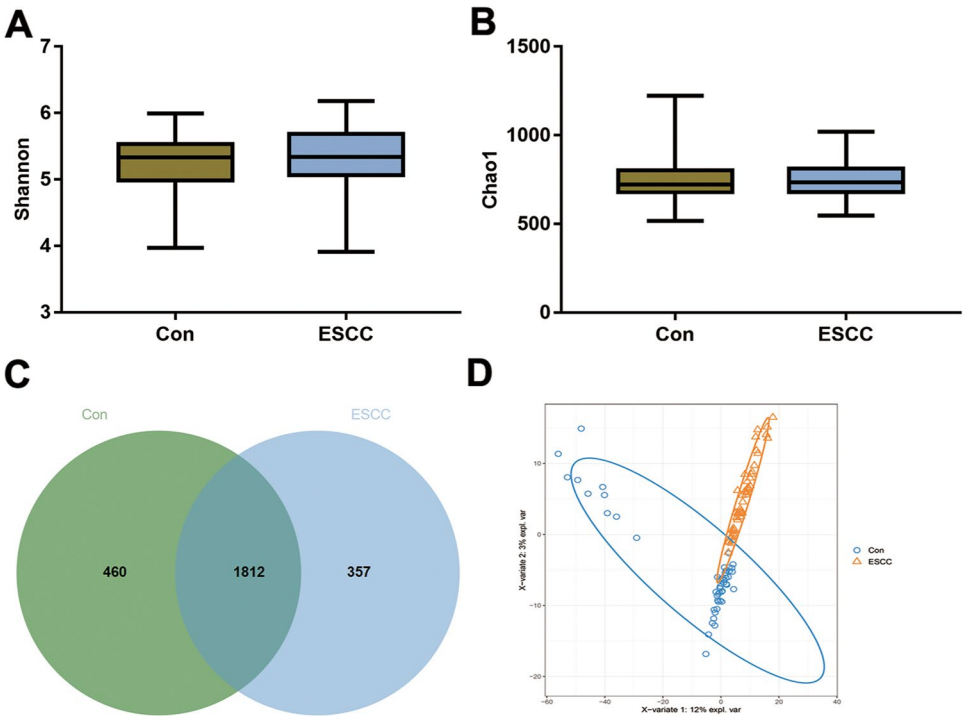


Fig. 2 The microbial α diversity and β diversity analysis in different groups. **A** Chao 1 index was higher in ESCC group than in Con group. **B** The ESCC group had a slightly higher Shannon index, in comparison with ConT group. **C** A Venn diagram displayed the overlaps between the two groups. **D** PLS-DA revealed different microbial community structures in the two groups. $*p < 0.05$

each group. *Bacteroidota*, *Firmicutes*, *Proteobacteria*, *Fusobacteria*, *Actinobacteria*, and *Patescibacteria* were the six dominant bacteria phyla of oral flora in the two groups.

The oral flora of normal esophageal was composed mainly by *Bacteroidota* (33.8%), *Firmicutes* (21.9%), *Proteobacteria* (25.6%), *Fusobacteria* (12.0%), *Actinobacteria* (3.5%), and *Patescibacteria* (2.2%), plus another ~ 1% of unidentified bacteria. At the genus level, *Prevotella* (24.9%) was the main contributor to the microbiota profile, followed by *Neisseria*, *Veillonella*, and *Streptococcus* (17.9%, 8.2% and 8.7%, respectively); other subdominant genera were *Leptotrichia (4.6%)*, *Fusobacterium (7.4%)*, *Alloprevotella (5.2%)*, *Aggregatibacter (4.8%)*, *Porphyromonas (3.0%)*, *Haemophilus (2.3%)*, *Rothia (1.7%)*, *Actinomyces (1.3%)*, *TM7x (1.0%)*, and *Absconditabacteriale (1.0%)* (Fig. 3A, D).

ESCCT group showed a significant decrease of *Proteobacteria* ($p = 0.0003$) together with a statistically significant robust increase of *Fusobacterioa* ($p = 0.0073$) at phylum level as compared to ConT groups. The higher relative abundance of *Fusobacteriales* was also found in ESCCT group compared with ConT groups at order level ($p = 0.0073$). ESCCT group also displayed a striking reduction in its microbial composition, such as *Neisseriaceae* ($p = 0.0003$) and *Pasteurellaceae* ($p = 0.0008$), in comparison with ConT groups at family level. When compared with ConT groups, decreased *Neisseria* ($p = 0.0027$) and *Aggregatibacter* ($p = 0.0030$) with overexpressed *Leptotrichia* ($p = 0.0001$) in ESCCT groups were identified at genus level (Fig. 3A–D).

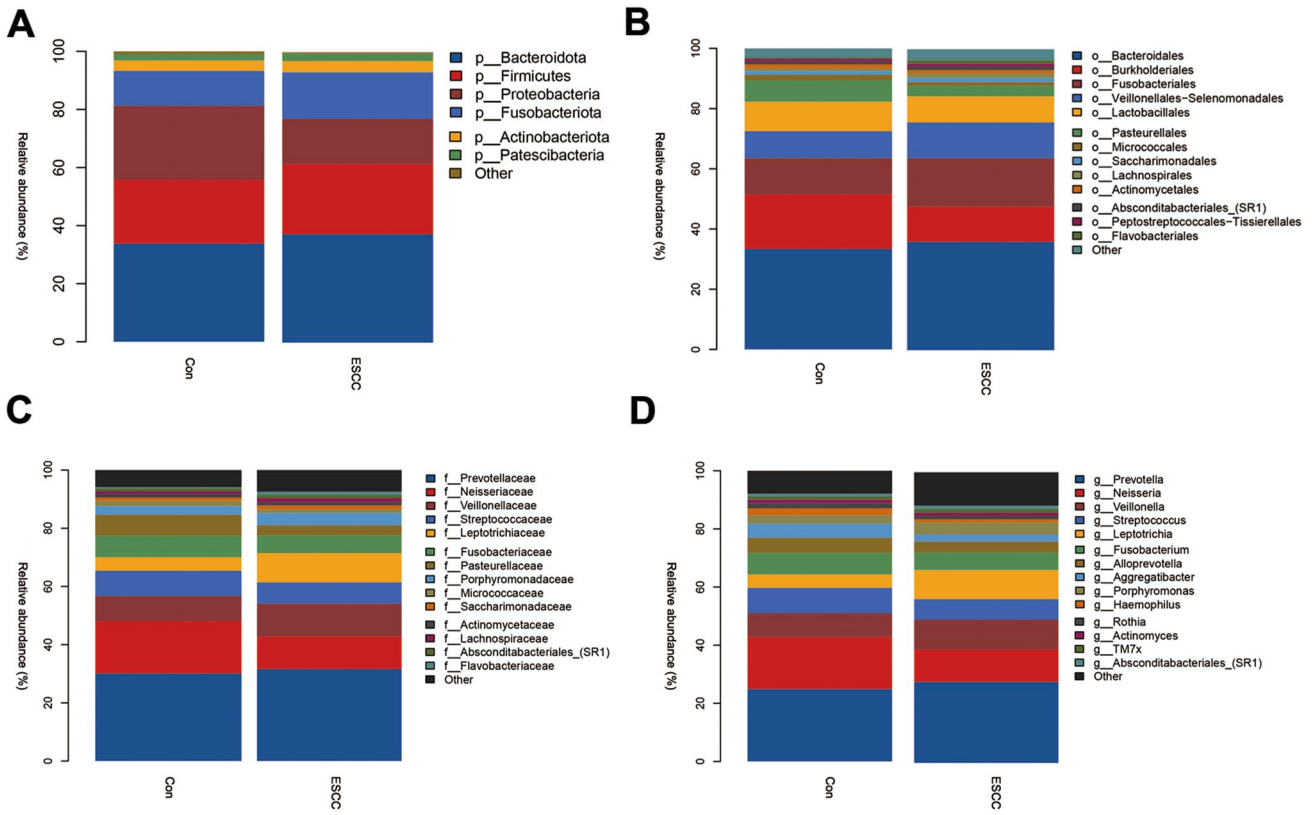


Fig. 3 Comparison of relative abundance between the two groups. **A** A bar plot of the relative abundance of the main bacterial taxa at phylum for ConT and ESCCT groups. **B** A bar plot of the relative abundance of the main bacterial taxa at order for ConT and ESCCT groups. **C** A bar plot of the relative abundance of the main bacterial taxa at family for ConT and ESCCT groups. **D** A bar plot of the relative abundance of the main bacterial taxa at genus level for ConT and ESCCT groups

Characterized oral microbial taxa associated with ESCC patients

As shown in Fig. 4, we examined the LDA coupled with LEfSe, and as a result we found that *Leptotrichiaceae*, *Leptotrichia*, *Fusobacteriales*, *Fusobacteria*, and *Fusobacteriota* were the key taxa contributing to the changes in the microbiome of ESCC patients. Thus, they could be detected as powerful markers in ESCCT groups. On the contrary, *Proteobacteria*, *Gammaproteobacteria*, *Neisseria* and *Neisseriaceae* were the key taxa resulting in the changes in the microbiome of healthy controls (Fig. 4).

Functional analysis of oral microbiota between groups

Finally, Phylogenetic Investigation of Communities by Reconstruction of Unobserved States (PICRUSt) was conducted to predict the metagenomes and identify the KEGG pathways involved in each group.

Patients with ESCCT showed an outstanding upregulation of microbial genes involved in translation and carbohydrate metabolism, while the characteristic of the signal transduction and drug resistance seems very poor, that is, compared to ConT groups (Fig. 5) antimicrobial were consistently downregulated.

Discussion

China is a country with high incidence of esophageal squamous cell carcinoma in the world, including Shanxi, Guangdong, Hebei, Jiangsu, Sichuan and Henan province (Lin et al. 2015). But the prognosis after radical resection of early esophageal cancer is better than expected. Based on the above situation, in fact, most patients diagnosed with esophageal cancer are in the middle and late stage, and a considerable number of patients have lost the opportunities of operation. Therefore, it is very important for us to improve the early diagnosis rate of esophageal cancer and the prevention research on the etiology of esophageal cancer. At present, the exact etiological mechanism of esophageal cancer is not clear.

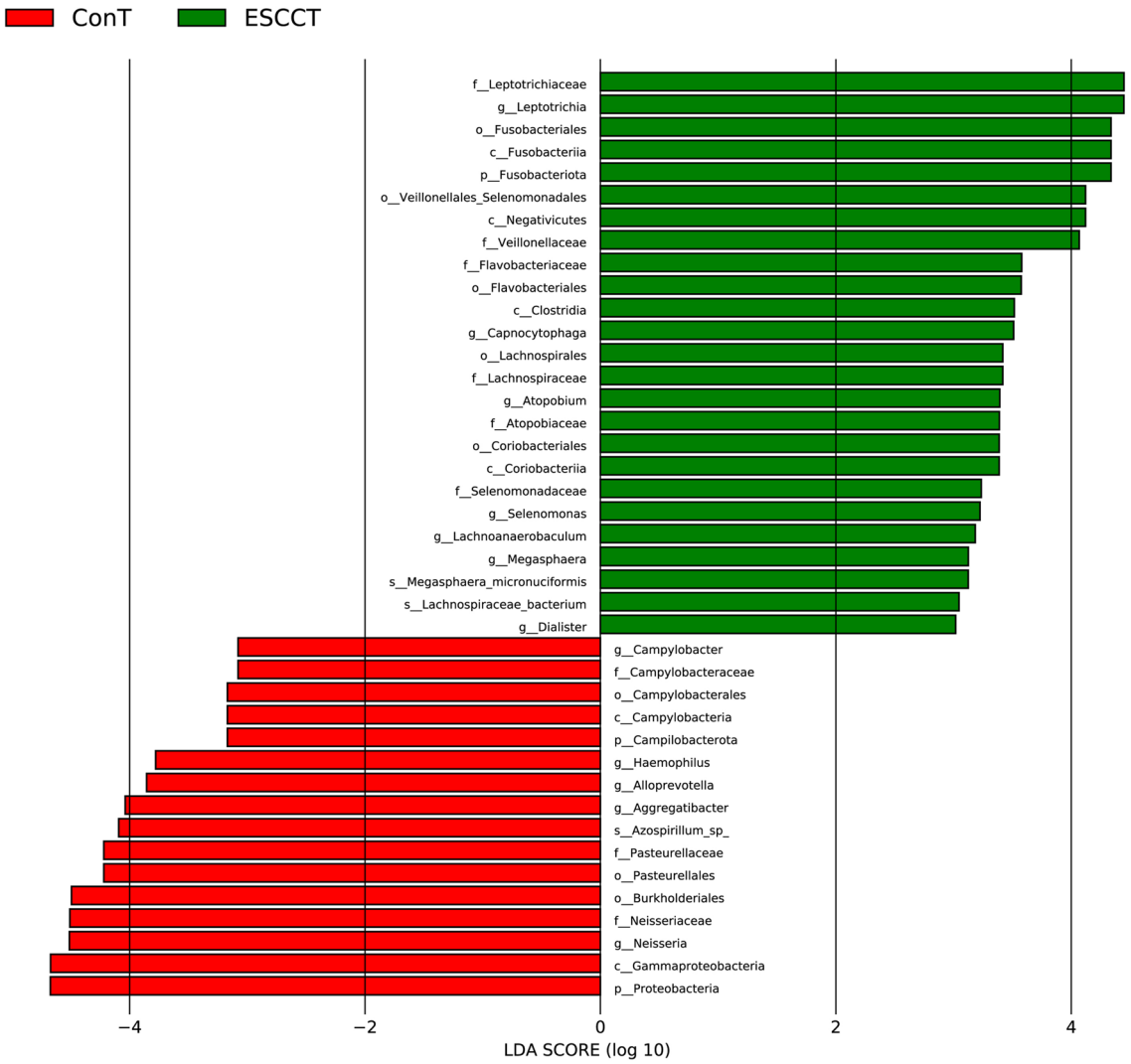


Fig. 4 LEfSe analysis showed the most abundant taxa from the phylum to the genus level between ConT and ESCCT groups

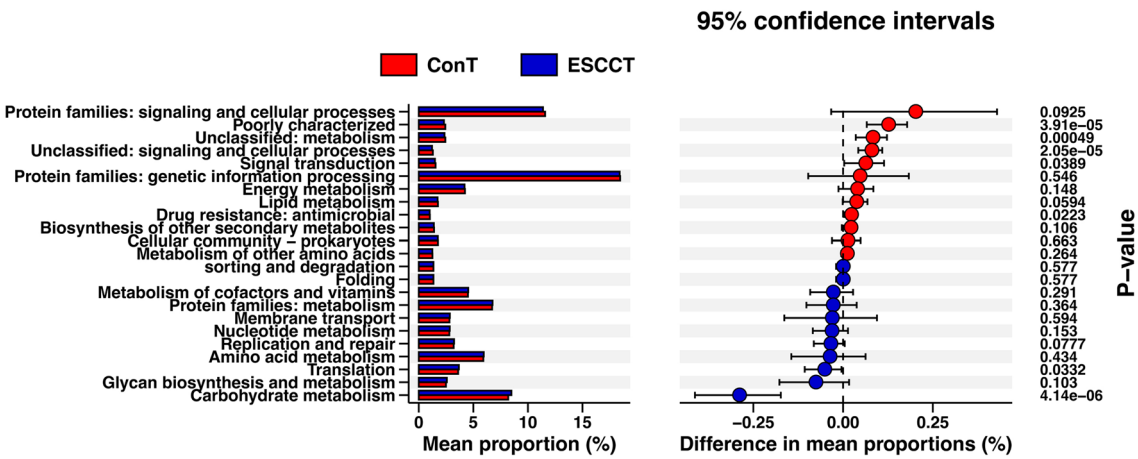


Fig. 5 The function prediction of the two groups. Differential KEGG pathways were analyzed using PICRUSt for the two groups. Significant differences between ConT and ESCCT groups were presented respectively

Increasing evidence has shown the crucial roles of bacterial infections in tumor development, including esophageal cancers (Garrett 2015). Although imbalanced microbiome has been well elucidated in EAC, microbial alterations of ESCC were still inconclusive. Recently, an ever-growing number of studies have demonstrated that the changes in the composition of our microbiome relate to numerous disease states, which raise the possibility that manipulation of these communities could be used to treat disease. With the implementation and progress of human microbiology project, bacterial dysbiosis in different sites on the human body was frequently found to be involved in the pathogenesis of several chronic diseases, such as diabetes, coronary heart disease, oral cavity and oropharyngeal tumors, (Guerrero-Preston et al. 2016; Schroeder and Backhed 2016; Lim et al. 2017).

Therefore, we used 16S rRNA gene sequencing to study the oral microorganisms of patients with ESCC, hoping to screen the bacteria specifically expressed in the oral cavity of the patients with ESCC, at the same time we found a higher oral microbiota richness and diversity without significant difference in ESCC groups than in healthy controls. It is obvious that the overall content and diversity of oral microbiota in ESCC patients have not changed significantly, which was supported by previous findings (Peters et al. 2017).

There was significantly increased proportional abundance for *Fusobacteriota* in ESCC patients, compared with healthy controls in Phylum level. *Fusobacteria* are Gram-negative anaerobic bacilli with species-specific reservoirs in the human mouth, gastrointestinal tract and elsewhere. An association between the presence of *F. nucleatum* and human colorectal cancer has emerged across both patient populations and disease stages. Currently, the *Fusobacteria* are divided into two families: the *Leptotrichiaceae*, which include the *Leptotrichia*, *Sneathia*, *Sebaldella* and *Streptobacillus* genera, and the *Fusobacteriaceae*, including *Psychrilyobacter*, *Ilyobacter*, *Propionigenium*, *Cetobacterium* and *Fusobacterium* genus. There was a high proportional abundance of *Leptotrichiaceae* in family level and *Leptotrichia* in genus level of ESCC patients. The characteristics and the role of *Leptotrichia* in human health remain unclear. *Leptotrichia*, in the first place, regarded as an opportunistic pathogen, which has a possibility to cause disease in the presence of local or general pre-disposing factors and it has been reported to be involved in various human infections (Card et al. 2015; Zhang et al. 2015), such as periodontitis, lung abscess, pneumonia, osteomyelitis, endocarditis, and bacteremia, in the second place, isolation of *Leptotrichia* species from infective endocarditis patients with normally functioning immune systems has also been reported (Higurashi et al. 2013; Kumagai et al. 2013; Schrimsher et al. 2013). Nevertheless, some studies have pointed out that *Leptotrichia* is associated with pancreatic cancer (Fan et al. 2018). In our study, we found that *Leptotrichia* may associated with ESCC, but in the end we need further human and experimental research to confirm this relationship.

In contrast to above-mentioned studies, we observed an inverse association of genus *Neisseria* with ESCC risk. Consistent with our findings, decreased relative abundance of *Neisseria* was found in esophageal cancer and lung cancer (Yan et al. 2015; Peters et al. 2017). *Neisseria* species are oral cavity commensals (Liu et al. 2015), and previous studies showed that oral *Neisseria* are depleted by cigarette smoking (Colman et al. 1976; Morris et al. 2013; Wu et al. 2016), a cause of ESCC. However, the deeper relationship between *Neisseria* and ESCC needs to be further studied.

In the study of Wang et al. (Wang et al. 2019), their LefSe analysis found that in the saliva sample, the high risk of ESCC may be related to Actinomyces and Atopobium, while the healthy control group is closely contacted with *Fusobacterium* and *Porphyromonas*. Our LEfSe analysis pointed out that *Proteobacteria*, *Gammaproteobacteria* and *Neisseria* were biomarkers for normal people while *Leptotrichiaceae, Leptotrichia*, *Fusobacteriales*, *Fusobacteria* and *Fusobacteriota* were biomarkers for ESCC patients.

Apart from compositional changes in bacterial taxa, we also predicted alterations in function across the groups. Otto Warburg reported that cancer cells consume tremendous amounts of glucose and metabolize the majority of the glucose into lactate, even in the presence of oxygen. This phenomenon is now referred to as aerobic glycolysis, or the Warburg effect, and it also represents a striking metabolic difference between cancer and most normal tissues. Substantial work has sought to target increased glycolysis including efforts to inhibit lactate production and excretion (Hamanaka and Chandel 2012; Doherty and Cleveland 2013; Hay 2016). In our study, we observed a rise of carbohydrate metabolism as well, which suggested a potential underlying mechanism of ESCC.

In the study, 16S rRNA gene sequencing was applied to evaluate the microbiome, with the constraints of the short read lengths acquired, sequencing mistakes, discrepancies resulting from the different areas targeted, and challenges in evaluating OTUs (Huse et al. 2010; Quince et al. 2011). Metagenomics and culturomics are reported as the better approaches for investigating oral microbiota (Martellacci et al. 2019; Patini 2020a, b). Metagenomics is made to investigate microbial organisms directly in their natural settings by sequencing their DNA, without the necessity for species separation and lab cultivation (Daubin et al. 2003; Martellacci et al. 2019). It allows for the study of all microorganisms, whether or not they can be cultivated, revealing the species present, and enabling the extraction of data on the functionality of microbial communities in their natural environments (Coughlan et al. 2015). However, metagenomics still has some limitations. It is unable to distinguish

between alive and dead bacteria, and to detect a population, bacteria must have a concentration of at least 10^5 cells per gram (Lagier et al. 2018). Compared with metagenomics, culturomics as a novel method for studying microbiota, could make up for the above deficiencies. Utilizing various culture conditions and identification using matrix-assisted laser desorption ionization-time of flight mass spectrometry (MALDI–TOF MS) make up the field of "culturomics" (Lagier et al. 2015). It enables the collection of live bacteria that can be examined for various microbiological characteristics as well as the culture and identification of previously unidentified bacteria. The first use of the culturomics technique for the investigation of the periodontal and periimplant bacteria in dentistry was described by Martellacci et al. The identification and cultivation of bacterial species never before isolated from the oral cavity were made possible by the study of the oral samples (Martellacci et al. 2020). Many systemic diseases have been reported associated with oral microbiota (Patini 2020a, b). Metagenomics was used in the majority of the research. The analysis cannot identify the rare bacterial species ($< 10^5$ cells per gram). Due to the high species diversity and frequently low amount of plaque samples, this affects genetic studies in dentistry. The issue could be resolved by culturomics, which enables the identification of bacterial populations at concentrations as low as 10^2 cells per gram. Culturomics would be a strong supplementary of metagenomics when studying microbiome. The combine use of metagenomics and culturomics would deepen our knowledge of oral microbiota and its associations with systemic diseases.

Conclusion

In summary, our data provided the evidence that some oral microbes may increase the risk of ESCC, while the other bacteria may have connection with the decreased risk. Clarification of these associations can be instructive in prevention strategy by eradication or inoculation in high-risk individuals. Prospective studies followed by animal experiments are needed to clarify their causal relationships.

Author contributions ZJ, ZZ and SW conceived, organized and supervised the project, and proofread the manuscript. ZJ and JW collected and analyzed the data. ZJ and XQ drafted the manuscript. XQ supervised statistical analysis. All authors approved the final version of the manuscript. All the authors critically revised and approved the final version.

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Availability of data and material All raw reads were stored in NCBI Sequence Read Archive (SRA) database, and the accession number is PRJNA853196.

Declarations

Competing interests The authors declare no competing interests.

Conflict of interest The authors have declared that no competing interest exists.

Ethical approval The study protocol was approved by the institutional review board of Nanjing Medical University, and all the experiments were performed in accordance with approved guidelines and regulations.

Consent to participate Informed consent was obtained from all individual participants included in the study.

Consent for publication Not applicable.

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